U-C-00125b(DSA-DM)
23 April 1968
SUPERSEDING
Int. Fed. Spec. U-C-00125a(DSA-DM)
8 April 1964 and
INTERIM REVISION OF
Fed. Spec. U-C-125
14 May 1954

INTERIM FEDERAL SPECIFICATION

CARRIERS, AMALGAM, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification U-C-125.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental amalgam carriers suitable for carrying and delivering a plastic mass of amalgam into a tooth cavity.

1.2 Classification. Amalgam carriers shall be of the following types, styles and sizes, as specified (see 6.1):

Type I. Lever type.

Style A. Single ended.

Size 1. Regular barrel.
Size 2. Large barrel.

Style B. Double ended. One end with regular barrel and opposing end with large barrel.

Type II. Plunger type.

FSC 6520

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2. APPLICABLE DOCUMENTS

2.1 Specifications, standards and drawing. The following specifications, standards and drawing of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
QQ-N-281	Nickel - Copper - Alloy Bar, Plate, Rod, Sheet, Strip, Wire Forgings and Structural and Special Shaped Sections.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

20725	Carrier, Amalgam, Lever Type.

(Copies of Military specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I. Type I amalgam carrier components shall conform to the requirements listed in table I, when tested as specified in 4.3.5. All materials shall be in accordance with GG-I-526 for the applicable type and class, except the plunger which shall conform to QQ-N-281 for composition and mechanical properties only.

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Table I. Material (Type I Components)

Component	Class	Style A and B
Barrel	4	Type 416 steel, cold-drawn
Plunger	B	Class B, form 2,** cold-drawn No. 1 temper, of QQ-N-281.
Lever	4	Type 420 steel with .30/ .36 carbon*
Lever spring	4	Type 420 steel with .30/ .36 carbon*
Handle, Nose, and Plug	8	Free machining yellow brass, chrome/nickel copper plated (as drawn), or:
	3	type 304 stainless steel tubing and type 303 stainless steel rod or bar stock.
Screw	3	Type 303 steel

*Hardened, tempered, electrolytically polished, and stress-relieved to a Rockwell hardness of C47 to C52.

** Class B, form 2 metal shall be capable of being bent back on itself and crushed together without any fractures or crazing appearing around the bend.

3.1.2 Type II. Type II, amalgam carrier components shall conform to the requirements of GG-I-526 as listed in table II, except where otherwise specified.

Table II. Material (Type II Components)

Component	Material
Barrels, thumb piece, finger rest, stop washer, lock and gland nuts	Class 8, Alloy No. 270, Yellow brass of GG-I-526, or as an alternate nickel plating only
Piston, plunger, tip, and spring stop	Class 4, type 416 stainless steel of GG-I-526
Flexible shaft	Shall be made of wire conforming to the requirements of QQ-N-281
Spring	Shall be made of music wire suitable for the purpose intended

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3.2 Design and construction.

3.2.1 Type I, style A. Type I, style A, amalgam carrier shall be suitable for receiving, carrying, and delivering a plastic mass of amalgam into a tooth cavity by means of depressing a lever by finger pressure. Carrier shall consist of a barrel, plunger, lever, lever spring, nose, screw, and handle. All component parts shall be properly oriented to each other to provide for a smooth operating instrument, free from lateral excursions and binding of components. One end of the plunger shall fit the base of the barrel snugly so as to exert a cleansing or scouring action on the base. The other end of the plunger shall fit into the nose, and shall be held in correct position by means of a screw. The inner surface of the barrel shall be smoothly and properly finished to prevent adherence of amalgam and shall be so aligned that there shall be no freezing of plunger to barrel during normal operative dental procedures. The lever shall be shaped to straddle the plunger nose and distal end of the handle. The lever shall be force fitted on the plunger with the point of contact serving as the fulcrum of the lever during operation. Lever spring shall be clipped to the nose and held in correct position by means of a screw as specified herein. The spring shall be so designed and related to the other components that upon expelling a mass of amalgam, the spring shall return to rest. The nose shall be drive fitted and soldered into the distal end of the handle, to properly hold the spring and plunger in the handle. The proximal end of the hollow handle shall be sealed by means of a plug. Type I, style A amalgam carriers shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 20725 for the applicable size.

3.2.2 Type I, style B. Type I, style B amalgam carrier shall be as specified in 3.2.1, except that it shall be a double ended carrier equipped with a regular barrel on one end and a large barrel on the opposing end in accordance with DPSC Drawing 20725.

3.2.3 Type II. Type II, amalgam carrier shall be suitable for receiving, carrying, and delivering a plastic mass of amalgam. The amalgam carrier shall receive the plastic mass with the spring in free release and deliver the plastic amalgam by depressing the thumb piece, which in turn compresses the spring. The amalgam carrier shall consist of a large straight barrel, a small bent barrel, plunger, piston, tip, spring, flexible shaft, thumb piece, finger rest, stop washer, lock nut, and with or without gland nut. Style and design of components shall be as shown in figure 1.

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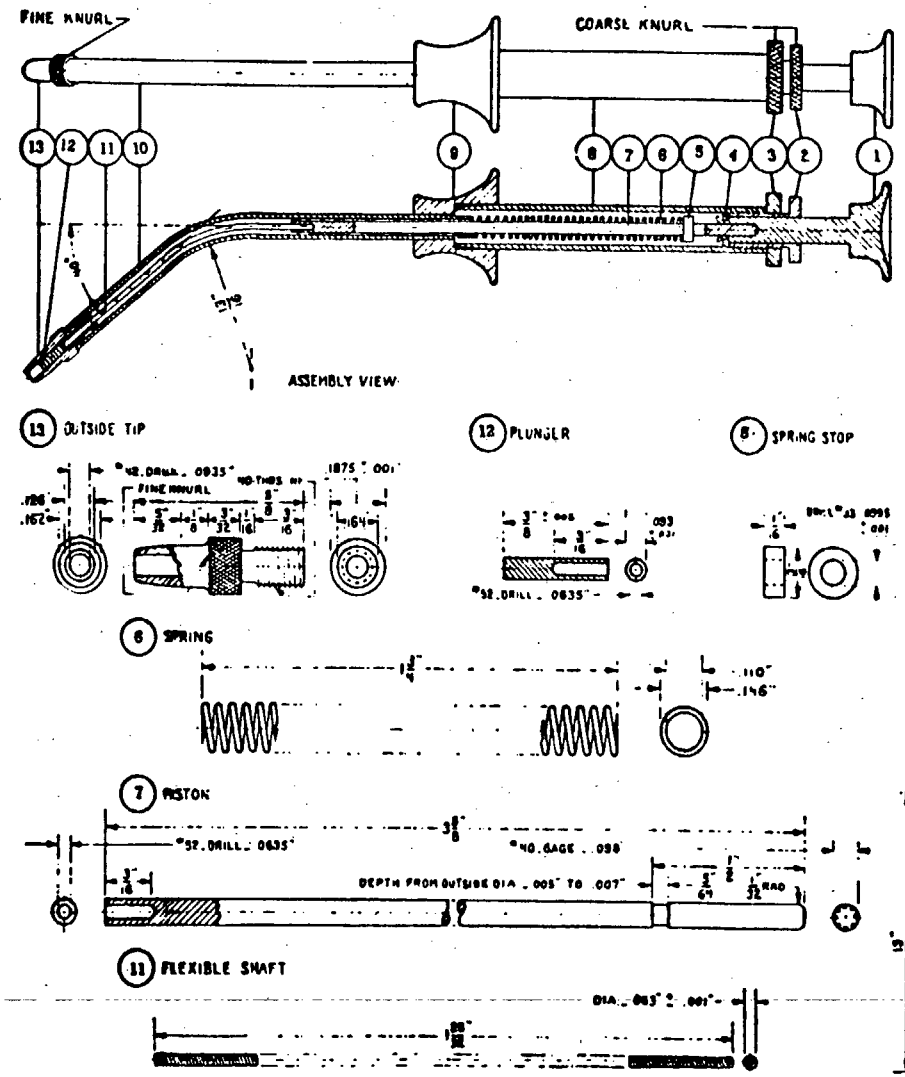


FIGURE 1

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3.3 Finish.

3.3.1 Type I. Type I, amalgam carrier shall be finished as specified in table III.

Table III. Finishes

Component	Finish
Barrel	Inside and outside of barrel shall have a smooth, polished finish
Plunger	Smooth, bright, polish
Lever	Bright polish or mirror
Lever Spring	Mirror or bright polish
Handle	Longitudinal scratch or longitudinal serrations
Nose and plug	Bright polish or plating
Screw	Bright finish

3.3.2 Type II. All parts of the type II amalgam carrier shall have a mirror finish, except for the handle which shall be knurled as shown in figure 1.

3.4 Corrosion tests.

3.4.1 Type I. The type I, amalgam carriers shall withstand the copper sulfate and boil tests specified in 4.3.2.

3.4.2 Type II. Component parts constructed of class 4, type 416 stainless steel shall withstand the copper sulfate and boil tests specified in 4.3.2. Component parts constructed of class 8, alloy number 270 yellow brass shall withstand the boil and salt spray tests specified in 4.3.2.

3.5 Operation test (for type I only). The amalgam carriers shall withstand 125,000 cycles of operation without breaking, permanent deformation, or impairment of proper alignment and operation, when tested as specified in 4.3.3.

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3.6 Identification marking. Each amalgam carrier shall be permanently and legibly stamped with the name or registered trademark of the manufacturer.

3.7 Workmanship. Workmanship shall be first class throughout. The amalgam carriers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables IV and V (see 4.2.1 and 4.2.2) and table VI (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table IV. Unit of product for examination shall be one amalgam carrier.

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Table IV. Sampling for Examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-1	1.5
Minor A	II	2.5
Minor B	S-1	4.0

4.2.2 Examination. The amalgam carriers shall be examined for defects including, but not limited to those listed in table V.

Table V. Classification of Defects

Categories	Defects
Critical	None Defined
Major A:	
101	Amalgam carrier not type, style, and size specified.
102	Component parts not free from cracks.
103	Edges and corners not free from burs or excessive sharpness.
104	Plunger fails to protrude from barrel when lever or thumb piece is depressed.
105	Carrier spring fails to return moving components to the same rest position.
106	Instrument fails to operate smoothly upon depressing and releasing lever or thumb piece.
107	Assembled components not properly fitted and aligned.
108	Plating on brass parts not adherent or not free from blisters, bare areas, and discoloration.
Major B:* (Applicable to type I only)	
151	Fit of plunger not within tolerance specified on drawing.
152	Inside diameter of barrel not within tolerance specified on drawing.
153	Inner surface of barrel not smooth, polished finish.
154	External surface of plunger engaging barrel not smooth, bright finish.
155	Thickness of spring not within tolerance shown on drawing.
156	Thickness of lever not within tolerance shown on drawing.
157	Narrowest section of the lever border around the oval shaped cutout less than 3/32 inch.

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Table V. Classification of Defects (continued)

Categories	Defects
Minor A:	
201	Finish (other than listed for Major B) not as specified.
202	Identification marking not present or not permanent and legible.
Minor B:*	
251	Dimensions (other than listed for Major B) not within tolerances shown on drawing (for type I only).

* Inspection for Major B and Minor B defects shall be conducted in process, or by disassembling the finished instrument when necessary.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table VI. The unit of product for test purposes shall be one amalgam carrier. The acceptance number shall be zero for all tests specified herein.

Table VI. Sampling for Tests

Characteristic	Requirements	Test procedure	Inspection level
Hardness	3.1.1 and 3.1.2	4.3.1	S-2
Copper sulfate test	3.4.1 and 3.4.2	4.3.2	S-2
Boil test	3.4.1 and 3.4.2	4.3.2	S-2
Salt spray test	3.4.2	4.3.2	S-2
Operation test	3.5	4.3.3	S-1
Plating test	3.1.2	4.3.4	S-1
Composition	3.1.1 and 3.1.2	4.3.5	*

* A sufficient amount of metal from each lot shall be selected for composition analysis.

4.3 Test methods.

4.3.1 Hardness test. Hardness shall be determined in accordance with method 243 (Rockwell) or method 244 (Diamond Pyramid) of FED-STD-151.

4.3.2 Corrosion tests. Copper sulfate, boil and salt spray tests shall be run in accordance with GG-I-526. Copper sulfate test, when applicable, shall be performed on the unassembled components.

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4.3.3 Operation test (type I only). This test shall be performed on type I carrier samples which were subjected to the boil test specified in 4.3.2. The carriers shall be held in a fixed horizontal position with the lever free end at its upper-most position. A wire shall be looped through both sides of the lever at plunger (barrel-end) cut-out area, and shall be tied in the direction of the expelling end, with a 1-pound weight suspended from the wire. The lever shall then be fully depressed and released at a uniform rate of 120 times per minute. The weight shall be removed and the carrier examined for characteristics specified in 3.5.

4.3.4 Plating tests. Plating tests for class 8, alloy number 270 material shall be run in accordance with GG-I-526, or as an alternate nickle plating only will be allowed.

4.3.5 Composition analysis. In the absence of certificates of quality for the raw material, composition analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One amalgam carrier, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VII shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VII. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1A-8.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
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Defense Supply Agency. - DM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION

U-C-00125b(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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